

Faculty of Science, Engineering and Built Environment

SEP105 Introduction to Programming for Engineers

Deakin University Unit Guide

Trimester 2, 2024

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Welcome

Engineers use computers for many different aspects of their day to day work. This might be to run simulations, perform data analysis, program a robot or embedded system, etc. These tasks, and others, often require an engineer to write custom programs. In this unit the fundamental concepts and skills of programming are introduced.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.
To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Benjamin Champion
Unit chair details

Campus: Geelong Waurn Ponds
Email: benjamin.champion@deakin.edu.au
Phone: +61 3 524 79683
Other members of the team and how to contact them

Details of any other staff will be provided on the unit site at the start of the trimester.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each unit. You are strongly encouraged to provide constructive feedback for this unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- Students liked that each topic of the unit was broken into smaller lectures.

The following aspects of the Unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- More examples of each topic have been added to the unit site.

If you have any concerns about the unit during the trimester, please contact the unit teaching team - preferably early in the

trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Your Unit Learning Outcomes

Each unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Learning Outcomes (ULO) for this unit. At the completion of this unit, successful students can:	Deakin Graduate Learning Outcomes
ULO1	Use appropriate planning tools to express logic and flow of a program.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy GLO4: Critical Thinking GLO5: Problem solving
ULO2	Define and apply appropriate variables and use them in calculations and assignment operations in a programming environment.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy GLO4: Critical thinking GLO5: Problem solving
ULO3	Define and create the logical control structures of sequence, selection and repetition and their combined use to express any logic in a programming environment.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy GLO4: Critical Thinking GLO5: Problem Solving
ULO4	Define and apply simple I/O methods using simple console and file inputs in a programming environment.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital Literacy GLO4: Critical Thinking GL05: Problem Solving
ULO5	Demonstrate communication skills of engineering methods in programming.	GLO3: Digital Literacy GLO2: Communication

These Unit Learning Outcomes are applicable for all teaching periods throughout the year.

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible for a pass in this unit, students must achieve a minimum of 50% in the presentation element of Project 3.

Brief summary of the hurdle requirements	Rationale
To be eligible for a pass in this unit, students must achieve a minimum of 50% in the presentation element of Project 3.	The hurdle is required for students to achieve ULO5 (demonstrate communication skills of the methods used in this work in this unit). It also ensures authenticity in students work by allowing assessment of their understanding, which is not achievable through software submissions or other tools such as turnitin alone.

Summative assessment (tasks that will be graded or marked)

NOTE: It is your responsibility to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you *may* be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59pm AEDT/AEST.

- Summative assessment task 1

Details	**Weekly Tests**
Brief description of assessment task	An online test each week that students need to individually complete covering class content up to the end of the current week.
Detail of student output	Students will be required to individually submit an online test on the content covered in the unit up to the current week. All tests will be submitted via the unit site.
Grading and weighting (% total mark for unit)	30% graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 – generate appropriate planning documentation ULO2 – choose and apply the appropriate variables ULO3 – choose and apply appropriate control structures to express the required logic ULO4 – use appropriate I/O methods for project
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - through completing quiz questions GLO3 - through participation in the online quiz GLO4 - through solving some quiz questions GLO5 - through solving some quiz questions
How and when you will receive feedback on your work	Tests will be marked, and students will receive feedback via the unit site within 2 weeks.
When and how to submit your work	Sunday of weeks 2 – 11 at 8pm (AEST) via the unit site

- Summative assessment task 2

Details	**Project 1**
Brief description of assessment task	Students will be required to individually write and submit a small program and associated documentation. Details will be provided on the unit site.
Detail of student output	Students will be required to individually submit an executable and the documented source code to the problem given on the unit site. Students will also be required to submit any other relevant documentation.
Grading and weighting (% total mark for unit)	15% graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 – generate appropriate planning documentation ULO2 – choose and apply the appropriate variables ULO3 – choose and apply appropriate control structures to express the required logic ULO4 – Use appropriate I/O methods for project
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - through writing and debugging a program GLO3 - through writing the program and associated documentation for the project GLO4 - through designing a solution to the given problem GLO5 - through designing and debugging the software for the given problem
How and when you will receive feedback on your work	The project will be marked against an assessment rubric and students will receive feedback via the unit site within 2 weeks.
When and how to submit your work	Sunday 4 August 2024 (Week 4) at 8pm (AEST) via the unit site.

- Summative assessment task 3

Details	**Project 2**

Brief description of assessment task	Students will be required to individually write and submit a small program and associated documentation. Details will be provided on the unit site.
Detail of student output	Students will be required to individually submit an executable and the documented source code to the problem given on the unit site. Students will also be required to submit any other relevant documentation.
Grading and weighting (% total mark for unit)	15% graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 – generate appropriate planning documentation ULO2 – choose and apply the appropriate variables ULO3 – choose and apply appropriate control structures to express the required logic ULO4 – Use appropriate I/O methods for project
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - through writing and debugging a program GLO3 - through writing the program and associated documentation for the project GLO4 - through designing a solution to a given problem GLO5 - through designing and debugging the software for the given problem
How and when you will receive feedback on your work	The project will be marked against an assessment rubric and students will receive feedback via the unit site within 2 weeks.
When and how to submit your work	Sunday 1 September 2024 (Week 7) at 8pm (AEST) via the unit site.

- Summative assessment task 4

Details	**Project 3**
Brief description of assessment task	Students will be required to individually write and submit a small program and associated documentation. Details will be provided on the unit site. Students will be required to make a presentation on the submitted work, which they must pass.
Detail of student output	Students will be required to individually submit an executable and the documented source code to the problem given on the unit site. Students will also be required to submit any other relevant documentation. Students will be required to give a short presentation on their solution, and answer questions about their project.
Grading and weighting (% total mark for unit)	40% marked and graded
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 – generate appropriate planning documentation ULO2 – choose and apply the appropriate variables ULO3 – choose and apply appropriate control structures to express the required logic ULO4 – Use appropriate I/O methods for project ULO5 – Demonstrate communication skills of the methods used in this work
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 - through writing and debugging a program GLO2 - through required presentation GLO3 - through writing the program and associated documentation for the project GLO4 - through designing a solution to the given problem GLO5 - through designing and debugging the software for the given problem
How and when you will receive feedback on your work	The project will be marked against an assessment rubric and students will receive feedback via the unit site on the unit site within 2 weeks.
When and how to submit your work	Sunday 22 September 2024 (Week 10) at 8pm (AEST) via the unit site. Presentations will be held during week 11.

Your learning experiences in this Unit - and your expected commitment

Typical study commitment

Students will on average spend 150 hours over the trimester undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.
Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

1 x 2 hour practical experience (studio) per week, 1 x 1 hour lecture per week (pre-recorded).
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online and collaborative learning including 1 x 2 hour practical experience (studio) per week.
The learning and assessment activities for campus students includes watching lecture recordings, attending studio workshops, assessment tasks, reading and study time. For online students they include online learning activities, discussion boards, designated activities, assessment tasks, readings and study time. Online students will also have access to the pre-recorded lectures.

Students are also expected to spend time reading the practical resources provided and completing self-directed problem-based exercises and assessment tasks.

This unit is designed to give a general introduction to programming skills, concepts and planning techniques. In this unit the content will be delivered by weekly online videos and reinforced in weekly studio sessions.
Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.
Unit learning resources

Your unit learning resources are available in your unit site.
The texts and reading list for the unit can be found on the University Library via the link below: SEP105 Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.
Essential learning resources

There is no prescribed textbook for this unit.

Recommended learning resources

The following text book is highly recommended:

- C++ How to program (10ed), Global Edition

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 2 teaching period begins	Monday 8 July 2024
Census date	Thursday 15 August 2024
Intra-trimester break	Monday 12 August – Sunday 18 August
Trimester 2 teaching period ends	Friday 27 September 2024
Study period (end-of-unit assessment /examination preparation period)	Monday 30 September – Friday 4 October 2024
End-of-unit assessment and examination period	Monday 7 October-Friday 18 October 2024
Inter-trimester break (the period between trimesters)	Monday 21 October-Friday 1 November 2024
Unit results released	Friday 1 November 2024 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Assessment activity
1	8 July 2024	Planning a Program	
2	15 July	Variables and Console IO	Weekly online test
3	22 July	Operators and Decision Making	Weekly online test
4	29 July	Arrays and Loops	Weekly online test Project 1
5#	5 August	Pointers and Functions	Weekly online test
6	19 August	Classes	Weekly online test
7	26 August	File IO	Weekly online test Project 2
8	2 September	Libraries	Weekly online test
9	9 September	Advanced Programming	Weekly online test
10	16 September	Advanced Programming	Weekly online test Project 3
11^	23 September	Project Presentations	Weekly online test Project 3 presentations

Intra-trimester break: **Monday 12 August - Sunday 18 August 2024** (inclusive)

^ AFL Grand Final Eve public holiday (University closed): **Friday 27 September 2024**